

The Gas Laws

Answer key with worked solutions

A. Understanding

1. Particles and pressure.

Pressure comes from gas particles colliding with the container walls. In a smaller volume the same particles hit the walls **more often**, so the pressure is higher.

2. Why kelvin?

Volume and pressure are proportional to the **absolute** temperature, which starts at absolute zero (0 K). The Celsius scale starts at an arbitrary point, so ratios of Celsius temperatures are meaningless (and 0 °C would mean dividing by zero).

B. Simple gas laws

3. Boyle.

$$V_2 = \frac{P_1 V_1}{P_2} = \frac{(100 \text{ kPa})(5.0 \text{ L})}{250 \text{ kPa}} = \mathbf{2.0 \text{ L}}$$

The units of kPa cancel, leaving L.

4. Charles.

$$T_1 = 20 \text{ }^\circ\text{C} + 273.15 = 293.15 \text{ K}; T_2 = -30 \text{ }^\circ\text{C} + 273.15 = 243.15 \text{ K}$$

$$V_2 = V_1 \times \frac{T_2}{T_1} = 250 \text{ mL} \times \frac{243.15 \text{ K}}{293.15 \text{ K}} = \mathbf{207 \text{ mL}}$$

The units of K cancel, leaving mL.

5. Gay-Lussac.

$$T_1 = 15 \text{ }^\circ\text{C} + 273.15 = 288.15 \text{ K}; T_2 = 45 \text{ }^\circ\text{C} + 273.15 = 318.15 \text{ K}$$

$$P_2 = P_1 \times \frac{T_2}{T_1} = 220 \text{ kPa} \times \frac{318.15 \text{ K}}{288.15 \text{ K}} = \mathbf{243 \text{ kPa}}$$

The units of K cancel, leaving kPa.

C. Combined gas law

6. Combined.

$$T_1 = 25\text{ }^{\circ}\text{C} + 273.15 = 298.15\text{ K}; T_2 = 100\text{ }^{\circ}\text{C} + 273.15 = 373.15\text{ K}$$

$$\begin{aligned} V_2 &= \frac{P_1 V_1 T_2}{P_2 T_1} \\ &= \frac{(1.00\text{ atm})(4.00\text{ L})(373.15\text{ K})}{(2.50\text{ atm})(298.15\text{ K})} \\ &= \mathbf{2.00\text{ L}} \end{aligned}$$

The units of atm and K cancel, leaving L.

D. Ideal gas law

7. Pressure.

Volume is in L and the answer is wanted in atm, so use $R = 0.08206\text{ L}\cdot\text{atm}/(\text{mol}\cdot\text{K})$

$$\begin{aligned} P &= \frac{nRT}{V} \\ &= \frac{(0.250\text{ mol}) \left(0.08206 \frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}} \right) (300\text{ K})}{5.00\text{ L}} \\ &= \mathbf{1.23\text{ atm}} \end{aligned}$$

The units of mol, K and L cancel, leaving atm.

8. Carbon dioxide.

$$(a) T = 27\text{ }^{\circ}\text{C} + 273.15 = 300.15\text{ K}$$

Pressure is in atm, so use $R = 0.08206\text{ L}\cdot\text{atm}/(\text{mol}\cdot\text{K})$

$$\begin{aligned} n &= \frac{PV}{RT} \\ &= \frac{(1.50\text{ atm})(2.00\text{ L})}{\left(0.08206 \frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}} \right) (300.15\text{ K})} \\ &= \mathbf{0.122\text{ mol}} \end{aligned}$$

The units of atm, L and K cancel, leaving mol.

$$(b) M(\text{CO}_2) = 12.01\text{ g/mol} + 2(16.00\text{ g/mol}) = 44.01\text{ g/mol}$$

$$m = n \times M = 0.12181\text{ mol} \times 44.01\text{ g/mol} = \mathbf{5.36\text{ g}}$$

E. Challenge

9. Molar mass.

$$T = 0\text{ }^{\circ}\text{C} + 273.15 = 273.15\text{ K}$$

Pressure is in atm, so use $R = 0.08206\text{ L}\cdot\text{atm}/(\text{mol}\cdot\text{K})$

$$\begin{aligned} n &= \frac{PV}{RT} \\ &= \frac{(1.00\text{ atm})(0.560\text{ L})}{\left(0.08206\frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}}\right)(273.15\text{ K})} \\ &= 0.02498\text{ mol} \end{aligned}$$

$$M = \frac{m}{n} = \frac{1.00\text{ g}}{0.02498\text{ mol}} = \mathbf{40.0\text{ g/mol}}$$

The gas is **argon** (39.95 g/mol).